

## Algebra First-Year Exam, August 2025, Part 1

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Answer 4 questions. You should indicate which 4 problems you wish to be graded. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. Prove that there exist at least three nonisomorphic nonabelian groups of order 12.
2. Prove that for  $n \geq 3$  the alternating group  $A_n$  is generated by 3-cycles. (You may assume that  $S_n$  is generated by transpositions.)
3. Let  $p$  be a prime, and let  $G = \text{GL}(3, \mathbb{F}_p)$ , where  $\mathbb{F}_p$  denotes the field of  $p$  elements.

(a) Compute the order of  $G$ .

(b) Let

$$U = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in \mathbb{F}_p \right\}.$$

Prove that  $U$  is a Sylow  $p$ -subgroup of  $G$ .

(c) Find the commutator subgroup of  $U$ .

4. Let  $G$  be a group acting on a set  $X$ .
  - (a) For  $x \in X$ , define the orbit of  $x$  and the stabilizer of  $x$ .
  - (b) Assume that  $G$  is finite. Prove that the size of the orbit of  $x$  divides the order of  $G$ .
5. Let  $G$  be a group.
  - (a) Define the terms automorphism of  $G$ , inner automorphism of  $G$  and characteristic subgroup of  $G$ .
  - (b) Prove that the set of automorphisms of  $G$  is a group in which the subset of inner automorphisms forms a normal subgroup.
  - (c) Prove that if  $N$  is a normal subgroup of  $G$  and  $H$  is a characteristic subgroup of  $N$ , then  $H$  is normal in  $G$ .
6. Let  $G$  be a group of order  $675 = 3^3 5^2$ .
  - (a) Prove that  $G$  has a normal Sylow 5-subgroup.
  - (b) Prove that  $G$  has an element of order 15.